Chapter 3. Experiments with a Single Factor: The Analysis of Variance

Contents

1 What If There Are More Than Two Factor Levels? 3

2 Experimental Design Terminology 3

3 Motivating Example: Tensile Strength 3

4 The Single-Factor ANOVA Model 4

4.1 Data layout and notation . . . 4

4.2 Effects model and means model . . . 4

4.3 Hypotheses . . . 4

5 Partitioning the Total Sum of Squares 4

6 Expected Mean Squares (Elementwise Proof) 5

7 The ANOVA Table and the F-test 6

7.1 Hand-computation formulas . . . 7

8 Parameter Estimation by Least Squares 8

9 Matrix (Regression) Approach to the One-Way ANOVA 8

9.1 Design matrix . . . 8

9.2 Vector differentiation . . . 9

9.3 Least squares in matrix form . . . 9

9.4 The hat matrix . . . 9

9.5 Sums of squares as quadratic forms . . . 10

9.6 Distribution theory . . . 10

9.7 Moments of fitted values and residuals . . . 12

10 Model Adequacy Checking 12

11 Statistical Tests for Equality of Variance 13

11.1 Bartlett's test . . . 13

11.2 Hartley's test (balanced, $n_i = n$) . . . 14

11.3 Levene's test (robust) . . . 14

12 Post-ANOVA Comparison of Means 14

12.1 Family-wise error rate . . . 14

12.2 Contrasts 14
12.3 Scheffé's method (all contrasts) 15
12.4 Tukey's test (all pairs) 15
12.5 Fisher's least significant difference (LSD) 15
12.6 Duncan's multiple range test 16

13 Unbalanced Data **16**

14 Transformations to Stabilize Variance **17**

15 Nonparametric Method: Kruskal–Wallis Test **17**

16 Summary of Chapter 3 **18**

1 What If There Are More Than Two Factor Levels?

- The t-test does not directly apply when there are more than two levels.
- In practice there are often more than two levels of interest, or several factors of simultaneous interest.
- The **analysis of variance (ANOVA)** is the appropriate analysis engine. It was developed by R. A. Fisher in the early 1920s for agricultural experiments and is widely used today in industrial experiments.

2 Experimental Design Terminology

Definition 3.1

- **Block**: group of homogeneous experimental units.
- **Factor**: a controllable experimental variable thought to influence the response.
- **Level**: a specific value of a factor.
- **Repeat tests**: two or more observations with the same levels of all factors.
- **Replication**: repetition of an entire experiment.
- **Response**: outcome or result of an experiment.
- **Unit**: entity on which a measurement is made.

3 Motivating Example: Tensile Strength

A new synthetic fibre for shirts; response = tensile strength. Cotton content (wt %) can vary between 10–40%; some non-linearity is anticipated. Five levels (15, 20, 25, 30, 35%) are chosen and each is replicated 5 times, runs in random order (a CRD with $a = 5$, $n = 5$, $N = 25$).

Cotton %	Observations					Total $y_{i\cdot}$	Average $\bar{y}_{i\cdot}$
15	7	7	15	11	9	49	9.8
20	12	17	12	18	18	77	15.4
25	14	18	18	19	19	88	17.6
30	19	25	22	19	23	108	21.6
35	7	10	11	15	11	54	10.8
						$y_{\cdot\cdot} = 376$	$\bar{y}_{\cdot\cdot} = 15.04$

Questions:

(1) Does cotton weight percent change the mean tensile strength? (2) Is there an optimum level?

4 The Single-Factor ANOVA Model

4.1 Data layout and notation

$$y_{i\cdot} = \sum_{j=1}^{n} y_{ij}, \quad \bar{y}_{i\cdot} = \frac{y_{i\cdot}}{n}, \quad y_{\cdot\cdot} = \sum_{i=1}^{a}\sum_{j=1}^{n} y_{ij}, \quad \bar{y}_{\cdot\cdot} = \frac{y_{\cdot\cdot}}{N}, \quad N = an.$$

4.2 Effects model and means model

Definition 3.2: Single-factor (one-way) model, fixed effects

Effects model: $y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, \ldots, a;\ j = 1, \ldots, n,$
μ: overall mean; τ_i: ith treatment effect; $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$, so $y_{ij} \overset{\text{ind}}{\sim} N(\mu + \tau_i, \sigma^2)$.
Means model: $y_{ij} = \mu_i + \varepsilon_{ij}$, with $\mu_i = \mu + \tau_i$.

Derivation: where the constraint $\sum \tau_i = 0$ comes from

The overall mean is defined as the average of the treatment means:

$$\mu = \frac{1}{a}\sum_{i=1}^{a} \mu_i = \frac{1}{a}\sum_{i=1}^{a}(\mu + \tau_i) = \mu + \frac{1}{a}\sum_{i=1}^{a}\tau_i \quad \Longrightarrow \quad \sum_{i=1}^{a}\tau_i = 0.$$

So $\tau_i = \mu_i - \mu$ is the deviation of the ith treatment mean from the overall mean.

- **Fixed effects model**: the a treatments are specifically chosen; conclusions apply only to them; τ_i are unknown constants.
- **Random effects model**: the a treatments are a random sample from a larger population; τ_i are random variables (Ch. 9).

4.3 Hypotheses

$$\begin{aligned} &H_0 : \mu_1 = \mu_2 = \cdots = \mu_a \\ &H_1 : \mu_i \neq \mu_j \text{ for at least one pair } (i, j) \end{aligned} \quad \Longleftrightarrow \quad \begin{aligned} &H_0 : \tau_1 = \cdots = \tau_a = 0 \\ &H_1 : \tau_i \neq 0 \text{ for at least one } i. \end{aligned}$$

5 Partitioning the Total Sum of Squares

Motivation. With more than two treatments, pairwise t-tests multiply the error rate. ANOVA instead asks one question, whether the treatment means differ at all, by splitting the total variation into a part explained by treatments and a part due to noise.

Theorem 3.3: Fundamental ANOVA identity

$$\underbrace{\sum_{i=1}^{a}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{\cdot\cdot})^2}_{SS_T} = \underbrace{n\sum_{i=1}^{a}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2}_{SS_{\text{Treatments}}} + \underbrace{\sum_{i=1}^{a}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i\cdot})^2}_{SS_E}.$$

Proof [Supplement]. Add and subtract $\bar{y}_{i\cdot}$:

$$\begin{aligned} SS_T &= \sum_{i=1}^{a}\sum_{j=1}^{n}\left[(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})+(y_{ij}-\bar{y}_{i\cdot})\right]^2 \\ &= \sum_i\sum_j(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})^2+\sum_i\sum_j(y_{ij}-\bar{y}_{i\cdot})^2+2\sum_i\sum_j(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})(y_{ij}-\bar{y}_{i\cdot}). \end{aligned}$$

The first term does not depend on j, so $\sum_i\sum_j(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})^2 = n\sum_i(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})^2$. For the cross product, pull out the factor that does not depend on j:

$$\sum_{i=1}^{a}\sum_{j=1}^{n}(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})(y_{ij}-\bar{y}_{i\cdot}) = \sum_{i=1}^{a}(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})\underbrace{\sum_{j=1}^{n}(y_{ij}-\bar{y}_{i\cdot})}_{=y_{i\cdot}-n\bar{y}_{i\cdot}=0} = 0.$$

Hence $SS_T = SS_{\text{Treatments}} + SS_E$. ■

- $SS_{\text{Treatments}}$ measures differences *between* treatment means; a large value reflects large differences.
- SS_E measures variation *within* treatments (pure error).
- Degrees of freedom: $an-1 = (a-1)+a(n-1)$, i.e. $N-1=(a-1)+(N-a)$.
- Sums of squares cannot be compared directly; **mean squares** can:

$$MS_{\text{Treatments}} = \frac{SS_{\text{Treatments}}}{a-1}, \qquad MS_E = \frac{SS_E}{a(n-1)} = \frac{SS_E}{N-a}.$$

Derivation: decomposition in terms of estimates

With $\hat{\mu}=\bar{y}_{\cdot\cdot}$, $\hat{\tau}_i=\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot}$, the fitted model is $y_{ij}=\hat{\mu}+\hat{\tau}_i+e_{ij}$, so $e_{ij}=y_{ij}-\bar{y}_{i\cdot}$ and

$$y_{ij}-\hat{\mu}=\hat{\tau}_i+e_{ij} \quad\Longrightarrow\quad SS_T=\sum_i\sum_j\hat{\tau}_i^2+\sum_i\sum_j e_{ij}^2.$$

6 Expected Mean Squares (Elementwise Proof)

Motivation. To build a test we need to know what each mean square estimates. Under H_0 both estimate σ^2, while the treatment mean square is inflated by $\sum n\tau_i^2/(a-1)$ otherwise; this is what makes their ratio informative.

Theorem 3.4

$$\mathrm{E}(MS_E)=\sigma^2, \qquad \mathrm{E}(MS_{\text{Treatments}})=\sigma^2+\frac{n\sum_{i=1}^{a}\tau_i^2}{a-1}.$$

Proof of $\mathrm{E}(MS_E)=\sigma^2$ [Supplement]. Write SS_E through the within-treatment sample variances $S_i^2=\frac{1}{n-1}\sum_j(y_{ij}-\bar{y}_{i\cdot})^2$:

$$MS_E=\frac{SS_E}{a(n-1)}=\frac{1}{a(n-1)}\sum_{i=1}^{a}\left[(n-1)\frac{\sum_j(y_{ij}-\bar{y}_{i\cdot})^2}{n-1}\right]=\frac{1}{a(n-1)}\sum_{i=1}^{a}(n-1)S_i^2.$$

Since $\bar{y}_{i\cdot} = \mu_i + \bar{\varepsilon}_{i\cdot}$, we have $\mathrm{E}(\bar{y}_{i\cdot}) = \mu_i$ and $\mathrm{Var}(\bar{y}_{i\cdot}) = \sigma^2/n$, hence $\mathrm{E}(\bar{y}_{i\cdot}^2) = \sigma^2/n + \mu_i^2$. Then

$$\begin{aligned}\mathrm{E}(S_i^2) &= \frac{1}{n-1}\mathrm{E}\Big(\sum_{j=1}^{n} y_{ij}^2 - n\bar{y}_{i\cdot}^2\Big) = \frac{1}{n-1}\Big(\sum_{j=1}^{n}\mathrm{E}(y_{ij}^2) - n\mathrm{E}(\bar{y}_{i\cdot}^2)\Big)\\ &= \frac{1}{n-1}\Big(\sum_{j=1}^{n}(\sigma^2+\mu_i^2) - n\Big(\frac{\sigma^2}{n}+\mu_i^2\Big)\Big) = \frac{1}{n-1}(n-1)\sigma^2 = \sigma^2.\end{aligned}$$

Therefore $\mathrm{E}(MS_E) = \frac{1}{a(n-1)}\sum_{i=1}^{a}(n-1)\sigma^2 = \sigma^2$. ■

Proof of $\mathrm{E}(MS_{\text{Treatments}})$ [Supplement]. Using $\bar{y}_{i\cdot} = \mu+\tau_i+\bar{\varepsilon}_{i\cdot}$ and $\bar{y}_{\cdot\cdot} = \mu+\bar{\varepsilon}_{\cdot\cdot}$ (since $\sum\tau_i = 0$),

$$\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = \tau_i + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot}).$$

Then

$$MS_{\text{Treatments}} = \frac{1}{a-1}\sum_{i=1}^{a}\sum_{j=1}^{n}(\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})^2 = \frac{n}{a-1}\sum_{i=1}^{a}\big(\tau_i + (\bar{\varepsilon}_{i\cdot}-\bar{\varepsilon}_{\cdot\cdot})\big)^2,$$

$$\begin{aligned}\mathrm{E}\Big[\sum_{i=1}^{a}\big(\tau_i + (\bar{\varepsilon}_{i\cdot}-\bar{\varepsilon}_{\cdot\cdot})\big)^2\Big] &= \mathrm{E}\Big[\sum_i \tau_i^2 + \sum_i(\bar{\varepsilon}_{i\cdot}-\bar{\varepsilon}_{\cdot\cdot})^2 + 2\sum_i \tau_i(\bar{\varepsilon}_{i\cdot}-\bar{\varepsilon}_{\cdot\cdot})\Big]\\ &= \sum_i \tau_i^2 + \mathrm{E}\Big[\sum_i(\bar{\varepsilon}_{i\cdot}-\bar{\varepsilon}_{\cdot\cdot})^2\Big] + 0.\end{aligned}$$

(The cross term has mean 0 since $\mathrm{E}\bar{\varepsilon}_{i\cdot} = \mathrm{E}\bar{\varepsilon}_{\cdot\cdot} = 0$.) For the remaining term, note $\bar{\varepsilon}_{\cdot\cdot} = \frac{1}{a}\sum_i \bar{\varepsilon}_{i\cdot}$, so

$$\begin{aligned}\mathrm{E}\Big[\sum_{i=1}^{a}(\bar{\varepsilon}_{i\cdot}-\bar{\varepsilon}_{\cdot\cdot})^2\Big] &= \mathrm{E}\Big[\sum_{i=1}^{a}\bar{\varepsilon}_{i\cdot}^2 - a\bar{\varepsilon}_{\cdot\cdot}^2\Big] = \sum_{i=1}^{a}\mathrm{E}(\bar{\varepsilon}_{i\cdot}^2) - a\mathrm{E}(\bar{\varepsilon}_{\cdot\cdot}^2)\\ &= \sum_{i=1}^{a}\frac{\sigma^2}{n} - a\cdot\frac{\sigma^2}{an} = (a-1)\frac{\sigma^2}{n}.\end{aligned}$$

(Equivalently: $\bar{\varepsilon}_{1\cdot},\dots,\bar{\varepsilon}_{a\cdot}$ are iid with variance σ^2/n, and the sample variance of them is unbiased.) Therefore

$$\mathrm{E}(MS_{\text{Treatments}}) = \frac{n}{a-1}\Big[\sum_i \tau_i^2 + (a-1)\frac{\sigma^2}{n}\Big] = \sigma^2 + \frac{n\sum_{i=1}^{a}\tau_i^2}{a-1}.$$

■

- MS_E is always an unbiased estimator of σ^2.
- Under H_0 ($\tau_i \equiv 0$), $\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2$, so $MS_{\text{Treatments}}$ also estimates σ^2.
- Let $F_0 = MS_{\text{Treatments}}/MS_E$. Heuristically,

$$\mathrm{E}(F_0) \approx \frac{\mathrm{E}(MS_{\text{Treatments}})}{\mathrm{E}(MS_E)} = \frac{\sigma^2 + \frac{n\sum\tau_i^2}{a-1}}{\sigma^2} \overset{H_0}{=} 1.$$

  Under H_0 the ratio is close to 1; under H_1 it tends to exceed 1. Decision: fail to reject if $F_0 \approx 1$, reject if $F_0 \gg 1$.

7 The ANOVA Table and the F-test

Motivation. The ratio of the two mean squares is near 1 under H_0 and large otherwise; under normality it has an F distribution, which gives the exact test summarized in the ANOVA table.

Source	SS	DF	MS	F_0
Between treatments	$SS_{\text{Treatments}}$	$a-1$	$MS_{\text{Treatments}}$	$MS_{\text{Treatments}}/MS_E$
Error (within)	SS_E	$N-a$	MS_E	
Total	SS_T	$N-1$		

Reference distribution: $F_0 \overset{H_0}{\sim} F_{a-1,\,N-a}$ (proved in Section 9). Reject H_0 if $F_0 > F_{\alpha,a-1,N-a}$.

7.1 Hand-computation formulas

Theorem 3.5: Computing formulas

$$SS_T = \sum_i \sum_j y_{ij}^2 - \frac{y_{\cdot\cdot}^2}{N}, \qquad SS_{\text{Treatments}} = \frac{1}{n}\sum_i y_{i\cdot}^2 - \frac{y_{\cdot\cdot}^2}{N}, \qquad SS_E = SS_T - SS_{\text{Treatments}}.$$

Proof [Supplement]. $\sum_i \sum_j (y_{ij} - \bar{y}_{\cdot\cdot})^2 = \sum\sum y_{ij}^2 - 2\bar{y}_{\cdot\cdot}\sum\sum y_{ij} + N\bar{y}_{\cdot\cdot}^2 = \sum\sum y_{ij}^2 - 2N\bar{y}_{\cdot\cdot}^2 + N\bar{y}_{\cdot\cdot}^2 = \sum\sum y_{ij}^2 - y_{\cdot\cdot}^2/N$. Similarly, using $\sum_i \bar{y}_{i\cdot} = a\bar{y}_{\cdot\cdot}$, $n\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = n\sum_i \bar{y}_{i\cdot}^2 - 2n\bar{y}_{\cdot\cdot}\sum_i \bar{y}_{i\cdot} + na\bar{y}_{\cdot\cdot}^2 = \frac{1}{n}\sum_i y_{i\cdot}^2 - 2N\bar{y}_{\cdot\cdot}^2 + N\bar{y}_{\cdot\cdot}^2 = \frac{1}{n}\sum_i y_{i\cdot}^2 - \frac{y_{\cdot\cdot}^2}{N}$. ■

Example 3.6: Tensile strength ANOVA, line by line

$$\begin{aligned}
\sum\sum y_{ij}^2 &= 6292, \qquad \frac{y_{\cdot\cdot}^2}{N} = \frac{376^2}{25} = \frac{141376}{25} = 5655.04,\\
SS_T &= 6292 - 5655.04 = 636.96,\\
SS_{\text{Treatments}} &= \frac{49^2 + 77^2 + 88^2 + 108^2 + 54^2}{5} - 5655.04\\
&= \frac{2401 + 5929 + 7744 + 11664 + 2916}{5} - 5655.04\\
&= \frac{30654}{5} - 5655.04 = 6130.8 - 5655.04 = 475.76,\\
SS_E &= 636.96 - 475.76 = 161.20.
\end{aligned}$$

Source	SS	DF	MS	F_0
Cotton %	475.76	4	118.94	$118.94/8.06 = 14.76$
Error	161.20	20	8.06	
Total	636.96	24		

$F_0 = 14.76 > F_{0.05,4,20} = 2.87$ ($P = 9.13 \times 10^{-6}$). Reject H_0: cotton weight percent significantly affects mean tensile strength.

8 Parameter Estimation by Least Squares

Derivation

Minimize

$$S(\mu, \tau_1, \ldots, \tau_a) = \sum_{i=1}^{a}\sum_{j=1}^{n} \varepsilon_{ij}^2 = \sum_{i=1}^{a}\sum_{j=1}^{n} (y_{ij} - \mu - \tau_i)^2.$$

Setting partial derivatives to zero at $(\hat{\mu}, \hat{\tau}_i)$:

$$\left.\frac{\partial S}{\partial \mu}\right|_{\hat{\mu},\hat{\tau}_i} = -2\sum_{i=1}^{a}\sum_{j=1}^{n}(y_{ij} - \hat{\mu} - \hat{\tau}_i) = 0,$$

$$\left.\frac{\partial S}{\partial \tau_i}\right|_{\hat{\mu},\hat{\tau}_i} = -2\sum_{j=1}^{n}(y_{ij} - \hat{\mu} - \hat{\tau}_i) = 0, \qquad i = 1, \ldots, a.$$

These give the $a+1$ **normal equations**

$$\begin{cases} N\hat{\mu} + n\hat{\tau}_1 + n\hat{\tau}_2 + \cdots + n\hat{\tau}_a = y_{\cdot\cdot} \\ n\hat{\mu} + n\hat{\tau}_1 = y_{1\cdot} \\ \quad \vdots \\ n\hat{\mu} + n\hat{\tau}_a = y_{a\cdot} \end{cases}$$

The first equation is the sum of the other a, so the system has rank a: one more condition is needed. Imposing $\sum \hat{\tau}_i = 0$, the first equation gives $N\hat{\mu} = y_{\cdot\cdot}$, so

$$\boxed{\hat{\mu} = \bar{y}_{\cdot\cdot}, \qquad \hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}, \quad i = 1, \ldots, a.}$$

Note: $\hat{\mu}_i = \hat{\mu} + \hat{\tau}_i = \bar{y}_{i\cdot}$ is the same whatever constraint is used (it is *estimable*), and $\hat{\tau}_i - \hat{\tau}_j = \bar{y}_{i\cdot} - \bar{y}_{j\cdot}$ is also estimable.

Fitted values and residuals: $\hat{y}_{ij} = \hat{\mu} + \hat{\tau}_i = \bar{y}_{i\cdot}$, $e_{ij} = y_{ij} - \hat{y}_{ij} = y_{ij} - \bar{y}_{i\cdot}$.

9 Matrix (Regression) Approach to the One-Way ANOVA

Motivation. ANOVA is a linear model with indicator regressors. Writing it in matrix form explains the constraints on the effects, connects the F-test to the general linear hypothesis, and extends to unbalanced and multifactor designs.

9.1 Design matrix

Derivation

For $a = 3$, $n = 2$ and the constraint $\tau_3 = -\tau_1 - \tau_2$, write $\mathbf{y} = (y_{11}, y_{12}, y_{21}, y_{22}, y_{31}, y_{32})^T$ and

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \mathbf{X} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{bmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \mu \\ \tau_1 \\ \tau_2 \end{pmatrix}, \qquad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}).$$

Hence $\mathrm{E}(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$, $\mathrm{Var}(\mathbf{y}) = \mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$ and $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$.

9.2 Vector differentiation

Theorem 3.7: Two derivative rules

1. If $\mathbf{a}$ is a constant $n \times 1$ vector, $\dfrac{\partial\, \mathbf{a}^T\mathbf{y}}{\partial \mathbf{y}} = \mathbf{a}$.
2. If $\mathbf{A}$ is a constant $n \times n$ matrix, $\dfrac{\partial\, \mathbf{y}^T\mathbf{A}\mathbf{y}}{\partial \mathbf{y}} = \mathbf{A}\mathbf{y} + \mathbf{A}^T\mathbf{y}$ ($= 2\mathbf{A}\mathbf{y}$ if $\mathbf{A}$ is symmetric).

Proof [Supplement]. (1) $\mathbf{a}^T\mathbf{y} = a_1y_1 + \cdots + a_ny_n$, so the kth component of the gradient is $\partial(a_1y_1 + \cdots + a_ny_n)/\partial y_k = a_k$.

(2) $\mathbf{y}^T\mathbf{A}\mathbf{y} = \sum_i \sum_j A_{ij}y_iy_j$. Differentiating w.r.t. y_k, the terms with $i = k$ give $\sum_j A_{kj}y_j = (\mathbf{A}\mathbf{y})_k$ and the terms with $j = k$ give $\sum_i A_{ik}y_i = (\mathbf{A}^T\mathbf{y})_k$. ■

9.3 Least squares in matrix form

$$\begin{aligned} S(\boldsymbol{\beta}) = \boldsymbol{\varepsilon}^T\boldsymbol{\varepsilon} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) &= \mathbf{y}^T\mathbf{y} - \mathbf{y}^T\mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} \\ &= \mathbf{y}^T\mathbf{y} - 2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} \qquad \text{(a scalar equals its transpose).} \end{aligned}$$

By rules (1) and (2) (with $\mathbf{a} = \mathbf{X}^T\mathbf{y}$, $\mathbf{A} = \mathbf{X}^T\mathbf{X}$ symmetric),

$$\frac{\partial S}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0} \quad \Longrightarrow \quad \mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y},$$

and if $(\mathbf{X}^T\mathbf{X})^{-1}$ exists, $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$.

9.4 The hat matrix

Definition 3.8

Fitted values $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \mathbf{H}\mathbf{y}$, where $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ (**hat matrix**). Residuals $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$.

Theorem 3.9: Properties of H

1. $\mathbf{H}^T = \mathbf{H}$ and $\mathbf{H}\mathbf{H} = \mathbf{H}$ (symmetric idempotent: orthogonal projection onto $\Omega(\mathbf{X})$, the column space of $\mathbf{X}$).
2. $\mathbf{H}\mathbf{X} = \mathbf{X}$; in particular $\mathbf{H}\mathbf{1} = \mathbf{1}$ whenever $\mathbf{1}$ is a column of $\mathbf{X}$.
3. $\mathbf{I} - \mathbf{H}$ is symmetric idempotent and $(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}$.
4. $\mathrm{tr}(\mathbf{H}) = \mathrm{rank}(\mathbf{X}) = p$ ($p = a$ here), $\mathrm{tr}(\mathbf{I} - \mathbf{H}) = N - p$.

Proof [Supplement]. (1)

$$\mathbf{H}\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\underbrace{\mathbf{X}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}}_{=\mathbf{I}}\mathbf{X}^T = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T = \mathbf{H}.$$

(2) $\mathbf{H}\mathbf{X} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X} = \mathbf{X}$. Writing $\mathbf{X} = (\mathbf{1}, \mathbf{X}_{-1})$, $\mathbf{H}\mathbf{X} = (\mathbf{H}\mathbf{1}, \mathbf{H}\mathbf{X}_{-1}) = (\mathbf{1}, \mathbf{X}_{-1})$, so $\mathbf{H}\mathbf{1} = \mathbf{1}$.

(3) $(\mathbf{I}-\mathbf{H})^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H}^2 = \mathbf{I} - \mathbf{H}$; $(\mathbf{I}-\mathbf{H})\mathbf{X} = \mathbf{X} - \mathbf{X} = \mathbf{0}$.

(4) $\operatorname{tr}(\mathbf{H}) = \operatorname{tr}(\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T) = \operatorname{tr}((\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}) = \operatorname{tr}(\mathbf{I}_p) = p$ (cyclic property of trace). ■

Geometrically, $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto $\Omega(\mathbf{X})$ and $\mathbf{e} \in \Omega(\mathbf{X})^{\perp}$.

9.5 Sums of squares as quadratic forms

Let $\mathbf{J}_N = \mathbf{1}\mathbf{1}^T$ ($N \times N$ matrix of ones), so $\frac{1}{N}\mathbf{J}_N\mathbf{y} = \bar{y}_{\cdot\cdot}\mathbf{1}$.

$$SS_E = \sum_i \sum_j (y_{ij} - \bar{y}_{i\cdot})^2 = \sum_i \sum_j (y_{ij} - \hat{y}_{ij})^2 = \mathbf{e}^T\mathbf{e} = \mathbf{y}^T(\mathbf{I}-\mathbf{H})^T(\mathbf{I}-\mathbf{H})\mathbf{y} = \mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y},$$

$$SS_{\text{Treatments}} = \sum_i \sum_j (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = (\hat{\mathbf{y}}-\bar{\mathbf{y}})^T(\hat{\mathbf{y}}-\bar{\mathbf{y}}), \qquad \hat{\mathbf{y}} - \bar{\mathbf{y}} = \left(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\right)\mathbf{y},$$

$$= \mathbf{y}^T\left(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\right)\mathbf{y} \quad \text{(once we show } \mathbf{H} - \tfrac{1}{N}\mathbf{J}_N \text{ is idempotent).}$$

Theorem 3.10: $\mathbf{H} - \frac{1}{N}\mathbf{J}_N$ is symmetric idempotent

Proof [Supplement]. First, $\frac{1}{N}\mathbf{J}_N = \mathbf{1}(\mathbf{1}^T\mathbf{1})^{-1}\mathbf{1}^T$ is itself a projection (onto $\operatorname{span}\{\mathbf{1}\}$), so $\frac{1}{N^2}\mathbf{J}_N\mathbf{J}_N = \frac{1}{N}\mathbf{J}_N$.

① $\frac{1}{N}\mathbf{J}_N\mathbf{H} = \mathbf{1}(\mathbf{1}^T\mathbf{1})^{-1}\underbrace{\mathbf{1}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T}_{=(\mathbf{H}\mathbf{1})^T=\mathbf{1}^T} = \frac{1}{N}\mathbf{J}_N.$

② $\frac{1}{N}\mathbf{H}\mathbf{J}_N = \underbrace{\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{1}}_{=\mathbf{H}\mathbf{1}=\mathbf{1}}(\mathbf{1}^T\mathbf{1})^{-1}\mathbf{1}^T = \frac{1}{N}\mathbf{J}_N.$

Hence

$$\left(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\right)\left(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\right) = \mathbf{H}\mathbf{H} - \underbrace{\tfrac{1}{N}\mathbf{H}\mathbf{J}_N}_{②} - \underbrace{\tfrac{1}{N}\mathbf{J}_N\mathbf{H}}_{①} + \underbrace{\tfrac{1}{N^2}\mathbf{J}_N\mathbf{J}_N}_{=\frac{1}{N}\mathbf{J}_N}$$

$$= \mathbf{H} - \tfrac{1}{N}\mathbf{J}_N - \tfrac{1}{N}\mathbf{J}_N + \tfrac{1}{N}\mathbf{J}_N = \mathbf{H} - \tfrac{1}{N}\mathbf{J}_N.$$

■

Its rank is $\operatorname{tr}(\mathbf{H} - \frac{1}{N}\mathbf{J}_N) = a - \frac{1}{N}\cdot N = a - 1$.

9.6 Distribution theory

Theorem 3.11: chi-square of a quadratic form

Let $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ and $\mathbf{A}$ an $n \times n$ symmetric matrix of rank k. If $\sigma^2\mathbf{A}$ is idempotent, then

$$\mathbf{y}^T\mathbf{A}\mathbf{y} \sim \chi^2_{k,\,\lambda}, \qquad \lambda = \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{A}\mathbf{X}\boldsymbol{\beta} \quad \text{(noncentral; central if } \lambda = 0).$$

Theorem 3.12: independence of quadratic forms

Let $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, $\mathbf{A}, \mathbf{B}$ symmetric, with $\mathbf{y}^T\mathbf{A}\mathbf{y}$ and $\mathbf{y}^T\mathbf{B}\mathbf{y}$ both chi-squared. If $\mathbf{A}(\sigma^2\mathbf{I})\mathbf{B} = \mathbf{0}$, then $\mathbf{y}^T\mathbf{A}\mathbf{y} \perp\!\!\!\perp \mathbf{y}^T\mathbf{B}\mathbf{y}$.

Theorem 3.13: expectation of a quadratic form

If $\mathbf{y}$ is an $n\times 1$ random vector with $\mathrm{E}(\mathbf{y})=\boldsymbol{\mu}$, $\mathrm{Var}(\mathbf{y})=\mathbf{V}$, then $\mathrm{E}(\mathbf{y}^T\mathbf{A}\mathbf{y})=\mathrm{tr}(\mathbf{A}\mathbf{V})+\boldsymbol{\mu}^T\mathbf{A}\boldsymbol{\mu}$ (no normality needed).

Proof of C [Supplement]. A scalar equals its trace, and trace and expectation commute:

$$\begin{aligned}\mathrm{E}(\mathbf{y}^T\mathbf{A}\mathbf{y}) &= \mathrm{E}(\mathrm{tr}(\mathbf{y}^T\mathbf{A}\mathbf{y})) = \mathrm{E}(\mathrm{tr}(\mathbf{A}\mathbf{y}\mathbf{y}^T)) = \mathrm{tr}(\mathbf{A}\,\mathrm{E}(\mathbf{y}\mathbf{y}^T))\\ &= \mathrm{tr}(\mathbf{A}(\mathbf{V}+\boldsymbol{\mu}\boldsymbol{\mu}^T)) \qquad (\mathrm{Var}(\mathbf{y})=\mathrm{E}(\mathbf{y}\mathbf{y}^T)-\boldsymbol{\mu}\boldsymbol{\mu}^T)\\ &= \mathrm{tr}(\mathbf{A}\mathbf{V})+\mathrm{tr}(\mathbf{A}\boldsymbol{\mu}\boldsymbol{\mu}^T) = \mathrm{tr}(\mathbf{A}\mathbf{V})+\boldsymbol{\mu}^T\mathbf{A}\boldsymbol{\mu}.\end{aligned}$$

■

Theorem 3.14: $F_0\sim F_{a-1,N-a}$ under H_0

1. $SS_E/\sigma^2\sim\chi^2_{N-a}$ (central, always);
2. $SS_{\text{Treatments}}/\sigma^2\sim\chi^2_{a-1,\lambda}$ with $\lambda=n\sum_i\tau_i^2/\sigma^2$ (central under H_0);
3. $SS_E \perp\!\!\!\perp SS_{\text{Treatments}}$;
4. hence $F_0=\dfrac{SS_{\text{Treatments}}/(a-1)}{SS_E/(N-a)}\overset{H_0}{\sim}F_{a-1,N-a}$ and, in general, $F_0\sim F_{a-1,N-a,\lambda}$ (non-central).

Proof [Supplement]. ① $SS_E=\mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y}$. Let $\mathbf{A}=(\mathbf{I}-\mathbf{H})/\sigma^2$; then $\sigma^2\mathbf{A}=\mathbf{I}-\mathbf{H}$ is idempotent. By Theorem 3.11,

$$\frac{\mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y}}{\sigma^2}\sim\chi^2_{\mathrm{tr}(\mathbf{I}-\mathbf{H}),\ \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{I}-\mathbf{H})\mathbf{X}\boldsymbol{\beta}/\sigma^2},$$

with $\mathrm{rank}(\mathbf{I}-\mathbf{H})=\mathrm{tr}(\mathbf{I})-\mathrm{tr}(\mathbf{H})=an-a$ and noncentrality 0 since $(\mathbf{I}-\mathbf{H})\mathbf{X}=\mathbf{0}$. So $SS_E/\sigma^2\sim\chi^2_{an-a}$.

② $SS_{\text{Treatments}}=\mathbf{y}^T(\mathbf{H}-\frac{1}{N}\mathbf{J}_N)\mathbf{y}$; $\mathbf{A}=(\mathbf{H}-\frac{1}{N}\mathbf{J}_N)/\sigma^2$, $\sigma^2\mathbf{A}$ idempotent, $\mathrm{tr}=a-1$. The noncentrality is $\boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{H}-\frac{1}{N}\mathbf{J}_N)\mathbf{X}\boldsymbol{\beta}/\sigma^2$. Compute

$$\Big(\mathbf{H}-\tfrac{1}{N}\mathbf{J}_N\Big)\mathbf{X}=\mathbf{H}\mathbf{X}-\tfrac{1}{N}\mathbf{J}_N\mathbf{X}=\mathbf{X}-\mathbf{1}\Big(\tfrac{1}{N}\mathbf{1}^T\mathbf{X}\Big)=\mathbf{X}-(\mathbf{1},\mathbf{0},\ldots,\mathbf{0}),$$

because in a balanced design each coded column sums to zero ($\mathbf{1}^T\mathbf{X}=(N,0,\ldots,0)$). Therefore

$$\boldsymbol{\xi}:=\Big(\mathbf{H}-\tfrac{1}{N}\mathbf{J}_N\Big)\mathbf{X}\boldsymbol{\beta}=\mathbf{X}\boldsymbol{\beta}-\mu\mathbf{1}=(\underbrace{\tau_1,\ldots,\tau_1}_{n},\ldots,\underbrace{\tau_a,\ldots,\tau_a}_{n})^T,$$

and $\boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{H}-\frac{1}{N}\mathbf{J}_N)\mathbf{X}\boldsymbol{\beta}=\boldsymbol{\xi}^T\boldsymbol{\xi}=n\sum_i\tau_i^2$ (using idempotence). Under $H_0:\tau_i=0$, $\boldsymbol{\xi}=\mathbf{0}$.

③ By Theorem 3.12 with $\mathbf{A}=(\mathbf{I}-\mathbf{H})/\sigma^2$, $\mathbf{B}=(\mathbf{H}-\frac{1}{N}\mathbf{J}_N)/\sigma^2$:

$$\sigma^2\mathbf{A}\mathbf{B}=\sigma^2\cdot\frac{\mathbf{I}-\mathbf{H}}{\sigma^2}\cdot\frac{\mathbf{H}-\frac{1}{N}\mathbf{J}_N}{\sigma^2}=\frac{1}{\sigma^2}\Big(\mathbf{H}-\tfrac{1}{N}\mathbf{J}_N-\mathbf{H}\mathbf{H}+\tfrac{1}{N}\mathbf{H}\mathbf{J}_N\Big)=\frac{1}{\sigma^2}\Big(\mathbf{H}-\tfrac{1}{N}\mathbf{J}_N-\mathbf{H}+\tfrac{1}{N}\mathbf{J}_N\Big)=\mathbf{0}.$$

④ The ratio of two independent chi-squares, each divided by its df, is F:

$$F_0=\frac{MS_{\text{Treatments}}}{MS_E}=\frac{\dfrac{\mathbf{y}^T(\mathbf{H}-\frac{1}{N}\mathbf{J}_N)\mathbf{y}}{\sigma^2}\Big/(a-1)}{\dfrac{\mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y}}{\sigma^2}\Big/(N-a)}\sim F_{a-1,\,N-a,\,\boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{H}-\frac{1}{N}\mathbf{J}_N)\mathbf{X}\boldsymbol{\beta}/\sigma^2}.$$

■

Derivation: EMS via Theorem 3.13

With $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$, $\mathbf{V} = \sigma^2\mathbf{I}$:

$$\begin{aligned}
\mathrm{E}(SS_E) &= \mathrm{E}(\mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y}) = \mathrm{tr}((\mathbf{I}-\mathbf{H})\sigma^2\mathbf{I}) + \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{I}-\mathbf{H})\mathbf{X}\boldsymbol{\beta} \\
&= \sigma^2(\mathrm{tr}(\mathbf{I}) - \mathrm{tr}(\mathbf{H})) + 0 = \sigma^2(N-a), \\
&\Rightarrow\ \mathrm{E}(MS_E) = \mathrm{E}\Big(\frac{SS_E}{N-a}\Big) = \sigma^2. \\
\mathrm{E}(SS_{\text{Treatments}}) &= \mathrm{tr}\Big((\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N)\sigma^2\mathbf{I}\Big) \\
&\quad + \boldsymbol{\beta}^T\mathbf{X}^T\Big(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\Big)^T\Big(\mathbf{H} - \tfrac{1}{N}\mathbf{J}_N\Big)\mathbf{X}\boldsymbol{\beta} \\
&= \sigma^2(a-1) + \boldsymbol{\xi}^T\boldsymbol{\xi} = \sigma^2(a-1) + n\sum_{i=1}^{a}\tau_i^2, \\
&\Rightarrow\ \mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + \frac{n\sum_{i=1}^{a}\tau_i^2}{a-1}.
\end{aligned}$$

This reproduces the elementwise result.

9.7 Moments of fitted values and residuals

Theorem 3.15

Using $\mathrm{Cov}(\mathbf{Ax}, \mathbf{By}) = \mathbf{A}\mathrm{Cov}(\mathbf{x},\mathbf{y})\mathbf{B}^T$:

$$\begin{aligned}
\mathrm{E}(\mathbf{e}) &= (\mathbf{I}-\mathbf{H})\mathrm{E}(\mathbf{y}) = (\mathbf{I}-\mathbf{H})\mathbf{X}\boldsymbol{\beta} = (\mathbf{X}-\mathbf{X})\boldsymbol{\beta} = \mathbf{0}, \\
\mathrm{Var}(\mathbf{e}) &= \mathrm{Cov}((\mathbf{I}-\mathbf{H})\mathbf{y}, (\mathbf{I}-\mathbf{H})\mathbf{y}) = (\mathbf{I}-\mathbf{H})\sigma^2\mathbf{I}(\mathbf{I}-\mathbf{H})^T = \sigma^2(\mathbf{I}-\mathbf{H}), \\
\mathrm{Cov}(\mathbf{e}, \hat{\mathbf{y}}) &= \mathrm{Cov}((\mathbf{I}-\mathbf{H})\mathbf{y}, \mathbf{H}\mathbf{y}) = (\mathbf{I}-\mathbf{H})\sigma^2\mathbf{I}\,\mathbf{H}^T = \sigma^2(\mathbf{H}-\mathbf{H}\mathbf{H}) = \mathbf{0}, \\
\mathrm{Var}(\hat{\mathbf{y}}) &= \mathbf{H}\sigma^2\mathbf{I}\mathbf{H}^T = \sigma^2\mathbf{H}, \qquad \mathrm{E}(\mathbf{e}^T\mathbf{e}) = \mathrm{tr}(\sigma^2(\mathbf{I}-\mathbf{H})) = \sigma^2(N-a).
\end{aligned}$$

Under normality, $\mathrm{Cov}(\mathbf{e}, \hat{\mathbf{y}}) = \mathbf{0}$ means residuals and fitted values are independent — the basis of the "residuals vs fitted" plot. Note also $\sum_{\ell=1}^{N}\mathrm{Var}(\hat{y}_\ell) = \sigma^2\mathrm{tr}(\mathbf{H}) = a\sigma^2$.

Remark 3.16: SS decomposition in matrix form

$SS_T = \mathbf{y}^T(\mathbf{I} - \frac{1}{N}\mathbf{J}_N)\mathbf{y} = \mathbf{y}^T(\mathbf{H} - \frac{1}{N}\mathbf{J}_N)\mathbf{y} + \mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y} = SS_{\text{Treatments}} + SS_E$, with $\chi^2_{a-1,\lambda}$ and χ^2_{N-a} components ($\lambda = \boldsymbol{\xi}^T\boldsymbol{\xi}/\sigma^2$).

10 Model Adequacy Checking

Residuals $e_{ij} = y_{ij} - \hat{y}_{ij} = y_{ij} - \bar{y}_{i\cdot}$.

- **Normality**: normal probability (Q–Q) plot of residuals.
- **Constant variance**: residuals vs fitted values $\hat{y}_{ij}$; or formal tests (Bartlett, Levene).
- **Independence**: residuals vs time order of data collection.

Derivation: constructing a normal probability plot

Given $z_1, \dots, z_n$; is $z_i \sim N(\mu, \sigma^2)$?

1. Order: $z_{(1)} \le \cdots \le z_{(n)}$.
2. Consider a normal sample of size n with mean 0 and unit variance; let $u_{(1)} \le \cdots \le u_{(n)}$ be the mean values of the order statistics over (many) samples. Approximately
$$u_{(i)} \approx \Phi^{-1}\Big(\frac{i - 3/8}{n + 1/4}\Big) \quad \text{(Blom)}.$$
3. If z is normal, $z_{(i)} = \mu + \sigma \cdot$ (standard normal order stat), so $\mathrm{E}(z_{(i)}) = \mu + \sigma u_{(i)}$: the plot of $z_{(i)}$ vs $u_{(i)}$ is a straight line with intercept μ and slope σ.

11 Statistical Tests for Equality of Variance

$H_0 : \sigma_1^2 = \cdots = \sigma_a^2$ vs H_1: not all equal.

11.1 Bartlett's test

$$\chi_0^2 = 2.3026\,\frac{q}{c}, \qquad q = (N-a)\log_{10} S_p^2 - \sum_{i=1}^{a}(n_i - 1)\log_{10} S_i^2,$$
$$c = 1 + \frac{1}{3(a-1)}\Big(\sum_{i=1}^{a}(n_i - 1)^{-1} - (N-a)^{-1}\Big),$$

$S_p^2 = \sum_i (n_i - 1)S_i^2/(N-a)$. Reject H_0 if $\chi_0^2 > \chi_{\alpha,a-1}^2$.

Derivation: why q measures inequality

Let $A = \frac{1}{\sum_i (n_i - 1)}\sum_i (n_i - 1)S_i^2 = S_p^2$ (weighted *arithmetic* mean) and $B = \big[(S_1^2)^{n_1 - 1} \cdots (S_a^2)^{n_a - 1}\big]^{1/(N-a)}$ (weighted *geometric* mean). By the AM–GM inequality $A \ge B$, with equality iff all S_i^2 are equal. Then

$$(N-a)\log\frac{A}{B} = (N-a)(\log A - \log B) = (N-a)\log S_p^2 - \sum_i (n_i - 1)\log S_i^2 = q \ \ge 0,$$

and q is large when the sample variances are very different. ($2.3026 = \ln 10$ converts $\log_{10}$ to natural logs.)

Example 3.17: Tensile data

$S_i^2 = 11.2,\ 9.8,\ 4.3,\ 6.8,\ 8.2$; $S_p^2 = MS_E = 8.06$.

$$\begin{aligned} q &= 20\log_{10} 8.06 - 4(\log_{10} 11.2 + \log_{10} 9.8 + \log_{10} 4.3 + \log_{10} 6.8 + \log_{10} 8.2) \\ &= 20(0.9063) - 4(1.0492 + 0.9912 + 0.6335 + 0.8325 + 0.9138) = 18.127 - 17.681 = 0.446, \end{aligned}$$
$$c = 1 + \frac{1}{3\cdot 4}\Big(\frac{5}{4} - \frac{1}{20}\Big) = 1 + \frac{1.2}{12} = 1.10, \qquad \chi_0^2 = 2.3026\frac{0.446}{1.10} = 0.933.$$

$0.933 < \chi_{0.05,4}^2 = 9.49$: fail to reject equal variances.

11.2 Hartley's test (balanced, $n_i = n$)

$H = \max S_i^2 / \min S_i^2$; reject if $H > H(a, n; 1-\alpha)$. Tensile: $H = 11.2/4.3 = 2.60$.

Bartlett and Hartley require normality and are very sensitive to departures from it.

11.3 Levene's test (robust)

$$L = \frac{\sum_{i=1}^{a} n_i(\bar{z}_{i\cdot} - \bar{z}_{\cdot\cdot})^2/(a-1)}{\sum_{i=1}^{a}\sum_{j=1}^{n_i}(z_{ij} - \bar{z}_{i\cdot})^2/(N-a)}, \qquad z_{ij} = |y_{ij} - \bar{y}_{i\cdot}|,$$

i.e. a one-way ANOVA on z_{ij}; reject if $L > F_{\alpha,a-1,N-a}$. **Modified Levene**: $z_{ij} = |y_{ij} - \text{median}(y_{i\cdot})|$. Implementation:

(1) compute z_{ij};

(2) run a one-way ANOVA. (Tensile: P-value ≈ 1, fail to reject.)

12 Post-ANOVA Comparison of Means

If H_0 is rejected, we do not know *which* means differ: the **multiple comparisons** problem.

12.1 Family-wise error rate

Derivation

$\alpha_{FW} = P(\text{at least one type I error} \mid \text{all } H_{0i} \text{ true})$. If each of r tests is done at level α and the tests are independent,

$$P(\text{no family-wise error}) = (1-\alpha)^r \;\Rightarrow\; \alpha_{FW} = 1 - (1-\alpha)^r.$$

E.g. $r = 3$, $\alpha = 0.05$: $\alpha_{FW} = 1 - 0.95^3 = 0.1426 < r\alpha = 0.15$. In general (independence not needed, by Bonferroni/Boole) $\alpha_{FW} \le r\alpha$. **Bonferroni**: test each H_{0i} at α/r, i.e. reject H_{0i} if $p_i < \alpha/r$; then $\alpha_{FW} \le \alpha$.

Proof of $1-(1-\alpha)^r \le r\alpha$ [Supplement]. By induction: $r = 1$ is equality. If $(1-\alpha)^r \ge 1 - r\alpha$ then $(1-\alpha)^{r+1} \ge (1-r\alpha)(1-\alpha) = 1 - (r+1)\alpha + r\alpha^2 \ge 1 - (r+1)\alpha$. Without independence, Boole: $P(\cup_i E_i) \le \sum_i P(E_i) = r\alpha$. ■

With 5 treatments there are $\binom{5}{2} = 10$ pairwise intervals; naive 95% intervals have simultaneous confidence only $\ge 1 - 10(0.05) = 0.5$.

12.2 Contrasts

Definition 3.18

A **contrast** is $\Gamma = \sum_{i=1}^{a} c_i\mu_i$ with $\sum_i c_i = 0$. E.g. $\Gamma = \mu_1 + \mu_2 - \mu_3 - \mu_4$; $\Gamma = \mu_1 - \mu_4$ ($H_0 : \Gamma = 0 \iff \mu_1 = \mu_4$).

Estimator $C = \sum_i c_i\bar{y}_{i\cdot}$:

$$\mathrm{E}(C) = \sum_i c_i \mathrm{E}(\bar{y}_{i\cdot}) = \sum_i c_i\mu_i = \Gamma, \qquad \mathrm{Var}(C) = \sum_i c_i^2 \mathrm{Var}(\bar{y}_{i\cdot}) = \frac{\sigma^2}{n}\sum_i c_i^2 \quad (n_i = n).$$

With MS_E for σ^2, $\dfrac{C-\Gamma}{\sqrt{MS_E \sum c_i^2/n}} \sim t_{N-a}$, giving the $100(1-\alpha)\%$ CI

$$\sum_i c_i \bar{y}_{i\cdot} \pm t_{\alpha/2,N-a} \sqrt{\frac{MS_E}{n} \sum_i c_i^2}.$$

Note: $\sum c_i = 0$ is what makes μ cancel: $\sum c_i \mu_i = \sum c_i(\mu + \tau_i) = \sum c_i \tau_i$.

12.3 Scheffé's method (all contrasts)

Scheffé's method gives simultaneous $100(1-\alpha)\%$ intervals for *all* contrasts:

$$C \pm S_\alpha, \qquad S_\alpha = \sqrt{(a-1)F_{\alpha,a-1,N-a}} \sqrt{\frac{MS_E}{n} \sum_i c_i^2}.$$

Reject $H_0 : \Gamma = 0$ if $|C| > S_\alpha$. Since S_α tends to be large, Scheffé is **conservative** (tends not to reject).

Example 3.19: Scheffé on tensile data

$\Gamma_1 = \mu_1 + \mu_3 - \mu_4 - \mu_5$, $\Gamma_2 = \mu_1 - \mu_4$.

$$C_1 = 9.8 + 17.6 - 21.6 - 10.8 = -5.0, \quad C_2 = 9.8 - 21.6 = -11.8,$$
$$\sqrt{(a-1)F_{0.05,4,20}} = \sqrt{4(2.87)} = 3.39,$$
$$S_{0.05,1} = 3.39\sqrt{\tfrac{8.06}{5} \cdot 4} = 3.39(2.54) = 8.61,$$
$$S_{0.05,2} = 3.39\sqrt{\tfrac{8.06}{5} \cdot 2} = 3.39(1.80) = 6.09.$$

$|C_1| = 5.0 < 8.61$: Γ_1 not significant. $|C_2| = 11.8 > 6.09$: $\mu_1 \neq \mu_4$.

12.4 Tukey's test (all pairs)

Studentized range $q = \dfrac{\bar{y}_{\max} - \bar{y}_{\min}}{\sqrt{MS_E/n}}$ with distribution $q(a, f)$, f = error df. Two means differ significantly if

$$|\bar{y}_{i\cdot} - \bar{y}_{j\cdot}| > T_\alpha = q_\alpha(a,f)\sqrt{\frac{MS_E}{n}} \qquad \Big(\text{unbalanced: } \frac{q_\alpha(a,f)}{\sqrt{2}}\sqrt{MS_E\Big(\frac{1}{n_i}+\frac{1}{n_j}\Big)}\Big).$$

Overall error rate is exactly α for all pairwise comparisons (balanced case).

12.5 Fisher's least significant difference (LSD)

$$|\bar{y}_{i\cdot} - \bar{y}_{j\cdot}| > LSD = t_{\alpha/2,N-a}\sqrt{\frac{2MS_E}{n}} \qquad \Big(\text{unbalanced: } t_{\alpha/2,N-a}\sqrt{MS_E\Big(\frac{1}{n_i}+\frac{1}{n_j}\Big)}\Big).$$

Controls error per comparison only (least conservative).

12.6 Duncan's multiple range test

Least significant ranges $R_p = r_\alpha(p, f)\sqrt{MS_E/n}$, $p = 2, \ldots, a$. Rank the means ascending $\bar{y}_{(1)} < \cdots < \bar{y}_{(a)}$; compare $\bar{y}_{(a)} - \bar{y}_{(1)}$ with R_a, $\bar{y}_{(a)} - \bar{y}_{(2)}$ with R_{a-1}, …, $\bar{y}_{(a)} - \bar{y}_{(a-1)}$ with R_2; then $\bar{y}_{(a-1)} - \bar{y}_{(1)}$ with R_{a-1}, etc. (a difference of means p steps apart in rank is compared with R_p). Significant if the difference exceeds the corresponding R_p.

Example 3.20: All pairwise comparisons for the tensile data

$\sqrt{MS_E/n} = \sqrt{8.06/5} = 1.270$.

- Tukey: $T_{0.05} = q_{0.05}(5, 20)\sqrt{8.06/5} = 4.24(1.270) = 5.38$.
- LSD: $t_{0.025,20}\sqrt{2(8.06)/5} = 2.086(1.796) = 3.75$.
- Duncan: $r_{0.05}(p, 20) = 2.95, 3.10, 3.18, 3.25$ for $p = 2, \ldots, 5 \Rightarrow R_2 = 3.75$, $R_3 = 3.94$, $R_4 = 4.04$, $R_5 = 4.13$.

Ordered means: $\bar{y}_1 = 9.8 < \bar{y}_5 = 10.8 < \bar{y}_2 = 15.4 < \bar{y}_3 = 17.6 < \bar{y}_4 = 21.6$.

Pair	|diff|	Tukey (5.38)	LSD (3.75)	Duncan (R_p)
4 vs 1	11.8	*	*	$> R_5 = 4.13$ *
4 vs 5	10.8	*	*	$> R_4 = 4.04$ *
4 vs 2	6.2	*	*	$> R_3 = 3.94$ *
4 vs 3	4.0	–	*	$> R_2 = 3.75$ *
3 vs 1	7.8	*	*	$> R_4$ *
3 vs 5	6.8	*	*	$> R_3$ *
3 vs 2	2.2	–	–	$< R_2$
2 vs 1	5.6	*	*	$> R_3$ *
2 vs 5	4.6	–	*	$> R_2$ *
5 vs 1	1.0	–	–	$< R_2$

LSD and Duncan agree: only (1,5) and (2,3) are not different. Tukey, being more conservative, also fails to separate (3,4) and (2,5). Level 30% (treatment 4) gives the largest strength.

Remark 3.21: Conservativeness ranking

From hardest to reject H_0 (most conservative) to easiest to reject H_0 (most liberal):

$$\text{Scheffé} \succ \text{Tukey} \succ \text{SNK} \succ \text{Duncan} \succ \text{Fisher's LSD}.$$

13 Unbalanced Data

If treatment i has n_i observations, $N = \sum_i n_i$; only the SS formulas change:

$$SS_T = \sum_{i=1}^{a}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{\cdot\cdot})^2, \quad SS_{\text{Treatments}} = \sum_{i=1}^{a} n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2, \quad SS_E = \sum_{i=1}^{a}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i\cdot})^2,$$

with hand formulas $SS_T = \sum\sum y_{ij}^2 - y_{\cdot\cdot}^2/N$, $SS_{\text{Treatments}} = \sum_i y_{i\cdot}^2/n_i - y_{\cdot\cdot}^2/N$. (The cross product in the identity still vanishes because $\sum_j(y_{ij} - \bar{y}_{i\cdot}) = 0$ for each i.) With the weighted constraint $\sum n_i\tau_i = 0$, $\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + \sum_i n_i\tau_i^2/(a-1)$.

14 Transformations to Stabilize Variance

Derivation: delta-method derivation

Suppose $\mathrm{Var}(y_i) = w(\mu_i)$, $\mu_i = \mathrm{E}(y_i)$. Find h such that $\mathrm{Var}(h(y))$ is (approximately) constant. First-order Taylor expansion about μ:

$$h(y) = h(\mu) + (y-\mu)h'(\mu) + \cdots \;\Rightarrow\; \mathrm{Var}\big(h(y)\big) \approx h'(\mu)^2\, w(\mu) \overset{\text{set}}{=} K.$$

Hence $h'(\mu) = \dfrac{\sqrt{K}}{\sqrt{w(\mu)}}$, i.e. $h(\mu) \propto \displaystyle\int \frac{d\mu}{\sqrt{w(\mu)}}$.

- $\sigma_i^2 \propto \mu_i$ ($w(\mu) = \mu$, Poisson-like): $h(\mu) = \int \mu^{-1/2} d\mu = 2\sqrt{\mu} \Rightarrow y^* = \sqrt{y}$.
- $\sigma_i \propto \mu_i$ ($w(\mu) = \mu^2$): $h(\mu) = \int \mu^{-1} d\mu = \log \mu \Rightarrow y^* = \log y$.
- $\sigma_i \propto \mu_i^2$ ($w(\mu) = \mu^4$): $h(\mu) = \int \mu^{-2} d\mu = -1/\mu \Rightarrow y^* = 1/y$.

In general if $\sigma \propto \mu^\alpha$, use $y^* = y^{1-\alpha}$ ($\log y$ when $\alpha = 1$). Diagnose with a funnel shape in the residuals-vs-fitted plot.

15 Nonparametric Method: Kruskal–Wallis Test

For H_0: the a treatment distributions are identical (location: $\mu_1 = \cdots = \mu_a$) when normality fails.

1. Rank all N observations jointly in ascending order; replace y_{ij} by its rank R_{ij} (average ranks for ties).
2. Compute

$$H = \frac{1}{S^2}\Big[\sum_{i=1}^{a} \frac{R_{i\cdot}^2}{n_i} - \frac{N(N+1)^2}{4}\Big], \qquad S^2 = \frac{1}{N-1}\Big[\sum_{i=1}^{a}\sum_{j=1}^{n_i} R_{ij}^2 - \frac{N(N+1)^2}{4}\Big].$$

3. Reject H_0 if $H > \chi^2_{\alpha,a-1}$ (large-sample approximation).

Derivation: no-tie form

Without ties the ranks are $1, \ldots, N$; $\bar{R}_{\cdot\cdot} = \frac{N+1}{2}$ and

$$S^2 = \frac{1}{N-1}\Big[\sum_{k=1}^{N} k^2 - N\Big(\frac{N+1}{2}\Big)^2\Big] = \frac{1}{N-1}\Big[\frac{N(N+1)(2N+1)}{6} - \frac{N(N+1)^2}{4}\Big] = \frac{N(N+1)}{12},$$

so

$$H = \frac{12}{N(N+1)}\sum_{i=1}^{a} n_i(\bar{R}_{i\cdot} - \bar{R}_{\cdot\cdot})^2 = \frac{12}{N(N+1)}\sum_{i=1}^{a} n_i\Big(\bar{R}_{i\cdot} - \frac{N+1}{2}\Big)^2.$$

(The bracket: $\frac{N(N+1)(2N+1)}{6} - \frac{N(N+1)^2}{4} = \frac{N(N+1)\,[2(2N+1) - 3(N+1)]}{12} = \frac{N(N+1)(N-1)}{12}$.) And $\sum_i \frac{R_{i\cdot}^2}{n_i} - \frac{N(N+1)^2}{4} = \sum_i n_i(\bar{R}_{i\cdot} - \frac{N+1}{2})^2$, which is $SS_{\text{Treatments}}$ computed on the ranks.

Example 3.22: Tensile data

Rank sums $R_{i\cdot} = 27.5,\ 66,\ 85,\ 113,\ 33.5$ ($N = 25$, ties present). $\sum\sum R_{ij}^2 = 5510$, $N(N+1)^2/4 = 4225$:

$$S^2 = \frac{5510 - 4225}{24} = 53.54,$$
$$\sum_i \frac{R_{i\cdot}^2}{5} = \frac{27.5^2 + 66^2 + 85^2 + 113^2 + 33.5^2}{5} = \frac{26228.5}{5} = 5245.7,$$
$$H = \frac{5245.7 - 4225}{53.54} = 19.06 > \chi^2_{0.05,4} = 9.49 \quad (P = 0.00076).$$

Reject H_0 — same conclusion as the F-test.

16 Summary of Chapter 3

Item	Result
LSE	$\hat{\mu} = \bar{y}_{\cdot\cdot}$, $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, $\hat{y}_{ij} = \bar{y}_{i\cdot}$
SS	$SS_T = SS_{\text{Trt}} + SS_E$, df $N-1 = (a-1) + (N-a)$
EMS	$\text{E}MS_E = \sigma^2$, $\text{E}MS_{\text{Trt}} = \sigma^2 + n\sum\tau_i^2/(a-1)$
Matrix	$SS_E = \mathbf{y}^T(\mathbf{I} - \mathbf{H})\mathbf{y}$, $SS_{\text{Trt}} = \mathbf{y}^T(\mathbf{H} - \mathbf{J}/N)\mathbf{y}$
Test	$F_0 = MS_{\text{Trt}}/MS_E \sim F_{a-1,N-a}$ under H_0
Contrasts	$\sum c_i\bar{y}_{i\cdot} \pm t_{\alpha/2,N-a}\sqrt{MS_E\sum c_i^2/n}$